

TED UNIVERSITY

Faculty of Engineering

Department of Computer Engineering

PROJECT SPESIFICATION REPORT

Project Title: AlerTED - Offline AI-Powered Emergency Medical Triage and Decision Support System

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1. Introduction

In the event of an earthquake, flood or conflict the first person to respond to the injury is often not a doctor or a healthcare professional but an ordinary individual who happens to be at the scene. This person may suddenly face multiple casualties and must decide within the minute whom to assist first and what actions to take. For someone without medical training, making these decisions correctly is extremely difficult; yet every decision made can directly influence an injured person's chances of survival.

Complicating matters further are the fact that in such situations, communication is often one of the first things to be lost. During major disasters or in remote areas, internet and telephone connectivity may fail, making it possible to seek remote assistance or look up information online. Consequently, any effective solutions for such a scenario must be capable of functioning without the need for any external connectivity.

AlerTED was developed to address these two challenges. It is a mobile decision- support system that operates entirely offline, providing step-by-step guidance to individuals without medical training during the processes of triage and initial intervention. This section begins with a general overview of the project followed by a discussion of its scope and objectives.

1.1 Description

AlerTED is an AI-powered offline capable mobile decision- support application designed for emergency situations where internet connectivity is limited or unavailable. For example, it can be used in disasters, conflict zones or operations in remote areas. The application aims to provide users with decision support during basic triage assessments and initial medical interventions. Its primary target audience consists of individuals who lack formal medical training.

The project focuses on two key areas. They are vital signs indicating the condition of an injured person and observable signs of trauma, through clear, step by step questions the system gathers essential information about the individual’s status. It uses AI-Based image processing to analyze photographs it takes from mobile devices. This analysis identifies visible traumatic signs such as open wounds, burns or active bleeding and provides additional information regarding these findings.

The data collected from the user, combined with visual information, is processed by decision-support mechanism based on established triage protocols. This mechanism assists in determining the appropriate initial medical intervention. AlerTED guides the user through the necessary first aid procedures, prioritizing individuals who require the most urgent care.

Both the image analysis and the decision support processes are designed to operate entirely on the mobile device without requiring an internet connection. The application is not intended to provide definitive medical diagnoses or replace professional medical care. Instead, it aims to support users in situations where professional support is delayed or communication is interrupted. AlerTED is being developed as a research project and its performance will be evaluated under controlled conditions.

1.2 Constraints

1.2.1 Economic Constraints

Since AlerTED is being developed as students' graduation project, its budget is limited. Development will be carried out using free and open-source tools, publicly available data sets and the team members' own mobile devices. Free or accessible computing resources will be used for model training. No paid datasets, commercial AI services or professional medical consultancy services will be purchased. The application is intended to be designed so that it can be offered to end users free of charge. For this reason, components that would generate ongoing costs, such as server hosting or cloud services, are avoided. The choice of offline operation is also an advantage in this respect.

1.2.2 Environmental Constraints

The application will be used in harsh physical environments such as disaster areas and conflict zones. In these environments, conditions such as low light, smoke, dust, rain and extreme temperatures may degrade the quality of the camera image and, consequently, the accuracy of the image analysis. Therefore, the image processing component must be developed and tested with varying image quality and field conditions in mind. In addition, since access to electricity may be limited in these areas, the application must be designed to minimize battery consumption. The energy consumption of model training should also be kept limited by preferring lightweight architecture.

1.2.3 Social Constraints

The target users are individuals without medical training who are under high stress. Therefore, questions and instructions must be free of medical terminology, short and easy to understand. The interface should require few steps and be usable with one hand. The application should be usable by people of different ages and levels of technological literacy. The application will initially be available in Turkish, which is a constraint for users who speak other languages. In addition, users may over-rely on the application and delay seeking professional medical help. The application must direct the user to professional medical help in every case.

1.2.4 Political Constraints

Since the usage scenarios of the application include conflict zones and operational areas, the project must not be perceived as a tool that supports any party or serves military purposes. AlerTED is designed to prioritize casualties and provide first-aid instructions based solely on their medical condition, regardless of their identity or affiliation. This approach is consistent with the humanitarian principle of neutrality. Furthermore, since the application processes health data, the project will be evaluated within the framework of Turkish legislation, in particular the Personal Data Protection Law No. 6698 (KVKK).

1.2.5 Ethical Constraints

AlerTED does not make definitive medical diagnoses and does not replace professional healthcare services. This must be clearly communicated to the user within the application.

Camera images and casualty information are sensitive health data; they must not be transferred off the device and must not be stored longer than necessary. Ethical issues are discussed in detail in Section 1.3.

1.2.6 Health and Safety Constraints

Casualty safety is the most critical constraint of the project. An incorrect triage decision or wrong first-aid guidance may lead to serious consequences. Therefore:

- The decision-support mechanism must rely only on triage protocols accepted in the literature. First-aid steps must be consistent with reliable sources.
- When the confidence level is insufficient or the collected data is contradictory, the system should avoid making a definitive triage decision and should direct the user to perform further assessment and seek professional medical support.
- When image analysis is unreliable the user must be directed to the question-based assessment.
- For ethical and safety reasons, evaluation will be carried out under controlled conditions using test datasets and simulation scenarios.

1.2.7 Manufacturability Constraints

AlerTED is a software product and does not require physical manufacturing. However, it must be able to run on a wide range of devices. The application will be developed to run on mobile devices that meet the specified minimum hardware requirements. Image processing model must be compressed to fit the limited processing, memory and storage capacity of mobile devices and must be able to run with on-device inference frameworks. Differences in camera quality and hardware across devices may affect performance. The project must be completed within the two-semester graduation project schedule and the capacity of a 5-person team.

1.2.8 Sustainability Constraints

Triage protocols and first-aid guidelines may be updated over time. The application's decision-support rules and first-aid content must be designed in a modular structure that can be updated without changing most of the code. Since the application works offline, updates can only be installed when the device is connected to the internet. The training process must be documented so that the image processing model can be retrained with new data in the future. In the long term, wider adoption of the application will require collaboration with healthcare institutions and regulatory approval processes for medical software.

1.2.9 Technical Constraints

- **Offline Operation:** All functions, including image processing and decision support, must run entirely on the device without an internet connection.
- **Response Time:** Since time is critical in emergencies, analysis results must be produced within a few seconds on a mid-range device.

- **Data Access:** Labeled, publicly available image datasets of findings such as open wounds, burns and active bleeding are limited. Since there is no medical expert on the team, label accuracy depends on the existing datasets.

1.3 Professional and Ethical Issues

AlerTED aims to help people with little or no medical knowledge during emergency situations. Since incorrect assessments or instructions may put injured people at risk, patient safety is one of the main responsibilities of the project. The application will not make medical diagnoses. Instead, it will support users by offering triage assessments and first aid instructions based on established protocols. Users will also be informed that the application has limitations and cannot replace professional medical care.

The application may collect sensitive information, such as photos of injured people and details about their condition. For this reason, protecting personal data is an important part of the development process. All data will be processed and stored on the mobile device without requiring an internet connection. Security measures will also be considered to protect this information from unauthorized access. The project will take Türkiye's Personal Data Protection Law (KVKK) into account.

Another important issue is the reliability of the AI model. Poor lighting, unclear images, or differences in skin tones may affect the accuracy of image analysis. Therefore, the results produced by the model will only be used as additional information during the assessment. The development team will test the system, identify its limitations, and report the results accurately. AlerTED will be developed as a research prototype and tested under controlled conditions.

2. Requirements

2.1 Functional Requirements

- **FR-01:** The application shall allow the user to create a new emergency incident record.
- **FR-02:** The application shall allow more than one injured person to be registered within the emergency.
- **FR-03:** The application shall assign a unique identifier to each injured person within an emergency incident.
- **FR-04:** The application shall collect the structured user information required by the selected triage protocol for the triage assessment.
- **FR-05:** The application shall allow the user to capture images of one or more visible injuries for each casualty using the mobile device’s camera.
- **FR-06:** The application shall analyze visible injuries, such as visible bleeding, open wounds, or burns using an on-device AI model.

- **FR-07:** The application shall evaluate both user provided information and supported visual findings from image analysis.
- **FR-08:** The application shall produce priority recommendations for the injured person according to validated triage protocol.
- **FR-09:** The application shall present the user with intervention priorities based on the assessment results in multi-casualty incidents.
- **FR-10:** The application shall provide step by step first-response guidance for the injured person, based on pre-defined first aid protocols, after the triage assessments.
- **FR-11:** The application shall allow the user to reassess a previously registered casualty and update the casualty's triage priority recommendation.
- **FR-12:** The application shall clearly inform the user when image analysis does not provide reliable results.
- **FR-13:** The application shall store emergency and casualty assessments data locally without requiring an internet connection.
- **FR-14:** The application shall guide the user to contact professional emergency services.

2.2 Non-Functional Requirements

- **NFR-01:** The application shall perform all core functions, including casualty assessments, intervention prioritization, first-aid guidance, without requiring an internet connection.
- **NFR-02:** All image analysis and decision support processes shall be performed on the mobile device.
- **NFR-03:** The application shall perform image analysis efficiently without causing unnecessary delays in the emergency assessment process.
- **NFR-04:** The application shall provide a simplified user interface with clearly labeled controls, readable text, and step by step navigation suitable for use under stressful conditions.
- **NFR-05:** The application shall utilize the mobile operating system’s built in security mechanisms to reduce unauthorized access.
- **NFR-06:** The application shall work offline on mobile devices stably.
- **NFR-07:** Triage and first-aid protocols shall be structured so that they can be updated without redesigning the application.

3. Backlog

Task Name	Story	Sprint Ready	Priority	Status	Story Points	Assigned to Sprint
Sprint 1 — Scope & Specifications	—	—	**High**	**Planned**	**21**	—
Define project scope and objectives	No	Yes	High	Complete	3	Yes
Prepare project proposal	No	Yes	High	Complete	5	Yes
Prepare project specifications report	No	Yes	High	In Progress	8	Yes
Define initial functional and non-functional requirements	No	Yes	High	In Progress	5	Yes
Sprint 2 — Research & Analysis	—	—	**High**	**Planned**	**42**	—
Research and compare mass-casualty triage protocols	No	Yes	High	Not Started	8	Yes
Research first-aid guidelines and safety considerations	No	Yes	High	Not Started	5	Yes
Review injury image datasets and licensing	No	Yes	High	Not Started	8	Yes
Define casualty, multiple-injury and reassessment scenarios	Yes	Yes	High	Not Started	5	Yes
Prepare use-case and activity diagrams	No	No	High	Not Started	8	Yes
Prepare analysis report	No	No	High	Not Started	8	Yes
Sprint 3 — High-Level Design	—	—	**High**	**Planned**	**42**	—
Design overall system architecture	No	No	High	Not Started	8	Yes
Design triage rule flow and first-aid content structure	No	No	High	Not Started	8	Yes
Design local data model and assessment history	No	No	High	Not Started	5	Yes
Create mobile UI wireframes and navigation flow	Yes	No	Medium	Not Started	5	Yes
Define offline AI model and integration approach	No	No	High	Not Started	8	Yes
Prepare high-level design report	No	No	High	Not Started	8	Yes
Sprint 4 — Prototype & Final Review	—	—	**High**	**Planned**	**34**	—
Build limited clickable or working prototype	No	No	Medium	Not Started	13	Yes
Test key offline and multi-casualty scenarios	No	No	High	Not Started	8	Yes
Review safety limitations and document open risks	No	No	High	Not Started	5	Yes
Prepare final presentation and demonstration	No	No	High	Not Started	8	Yes

4. References

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